

G. Indra Charan Reddy

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in Indra Charan [🔗](#) | 📷 indracharan-png [🔗](#)

Education

- **San Francisco State University (SFSU)** San Francisco, California
Masters in Computer Science - 4.0/4.0 GPA (Sem-1) Aug. 2024 - Present
- **Indian Institute of Information Technology, Lucknow (IIITL)** Uttarpradesh, India
Bachelor's in Information Technology - CGPA: 7.47/10 Dec. 2020 - Jun. 2024

Work Experience

Research Assistant San Francisco, CA

San Francisco State University (Dept. of Computer Science)

- Researching advanced topological graph theory under Prof. Timothy Sun, focusing on computational visualization of dense graphs projected onto complex surfaces (e.g., manifolds, simplicial complexes).
- Designing Python-based tools to automate graph embeddings and surface deformations, reducing manual analysis time by 40% for high-density structures (10k+ nodes).
- Publishing results in progress: Contributed to a preprint on applications in mathematical computer science, with focus on NP-hardness in topological graph partitioning.

Academic Research

Research Analysis on QUESTO

- Produced a 10-page summary report on QUESTO, analyzing the tool's approach to enabling non-experts in customizing objective functions for ML classification tasks.
- Presented key findings on the problem addressed, solution provided, limitations, and future directions, offering a critical evaluation of QUESTO's usability and impact on interactive ML.
- Collaborated with Professor to examine applications and implications of QUESTO in democratizing machine learning customization.

Projects

Period Pal | [Link 🔗](#)

- Developed a web application using an RNN-based machine learning model to predict menstrual cycle dates with 85% accuracy.
- Integrated the machine learning model with a FastAPI back-end and MongoDB, improving prediction accuracy by 30%.
- Optimized back-end logic to manage user data, achieving a response time of under 200ms for over 1,000 records..
- TechStack: Machine Learning-RNN, ReactJs, Express, MongoDB, NodeJS, FastApi, ML-backend code [here 🔗](#)

Attendance Tracker | [Link 🔗](#)

- Built a deep learning-based attendance system using face recognition, achieving 95% accuracy with the **Dlib** model.
- Designed the front-end UI with ReactJS, improving interaction efficiency and the user experience.
- Developed a real-time attendance dashboard in NodeJS, reducing manual tracking time by 60%.
- TechStack: HTML, CSS, React, NodeJs, Python

Slot Machine Game | [Link 🔗](#)

- Implemented a user-interactive slot machine game in JavaScript, enabling real-time betting and spinning mechanics.
- Implemented a random symbol generation system using matrix manipulation techniques, simulating slot reels with a 100% randomness factor.
- Engineered a winnings calculation algorithm, ensuring a dynamic update of player balance with no lag time, reducing game processing delay by 20%.
- Tech Stack: JavaScript

Current Positions

- **Governing Gators Intern - Strategic Issues Committee** SFSU, San Francisco, CA
Board Member, Associated Students Oct, 2024 - Present
 - Represent over 27,000 students at SFSU by advocating for their interests, partnering with campus leadership, and developing 3+ essential student support services each month.
- **Executive Committee Member** SFSU, San Francisco, CA
International Education Exchange Council (IEEC) Sept, 2024 - Dec, 2024
 - Foster intercultural exchange for 1,500+ international and domestic students by organizing weekly events and collaborating with the Division of International Education.
- **Project Connect Intern** SFSU, San Francisco, CA
Associated Students Sept, 2024 - Dec, 2024
 - Support over 100 underrepresented students each semester by providing essential resources, fostering networking opportunities, and offering guidance for a smoother college transition.

ACHIEVEMENTS AND WORKSHOPS

- * **1506 rating** in Leetcode DarthVader986 [↗](#)
- * Reached 2nd place in the institute on InterviewBit [↗](#)
- * Solved 1000+ problems across GeeksForGeeks, CodeChef, CodeForces, InterviewBit, and Leetcode.
- * Secured top 5 in the **Samsung Innovation Campus Hackathon**, a 3-day event. Nov. 2023
- * Participated in **HackOfiesta'22** (Annual Hackathon conducted by IIIT Lucknow) Oct. 2022
- * Joined IIT Kanpur's **AI/ML Weather Workshop** Mar. 2022

SKILLS

- **Languages:** C, C++, SQL, HTML/CSS, Java, Java Script
- **Technologies:** Linux, VScode, IntelliJ, Github
- **Toolset:** Data Structures, Algorithms, Reactjs, Javascript, Nodejs, Express, HTML, CSS, Machine Learning, Deep Learning.
- **Course Structure:** DSA, Discrete Mathematics, Computer Networks, Operating System, OOPS, DBMS, Cryptography.
- **Soft Skills:** Teamwork, Time Management, Responsible, Consistent, Communication Skills, Adaptability